Capstone Challenge

2026-2027

Grade 12 Semester 1

Challenge Title: Information Technology and Artificial Intelligence (AI) Applications

Theme: Communication, Sensing, Information, Informatics

Egypt Grand Challenges

- Increase the industrial and agricultural base of Egypt.
- Public health issues and diseases.
- Improve the scientific and technological environment.

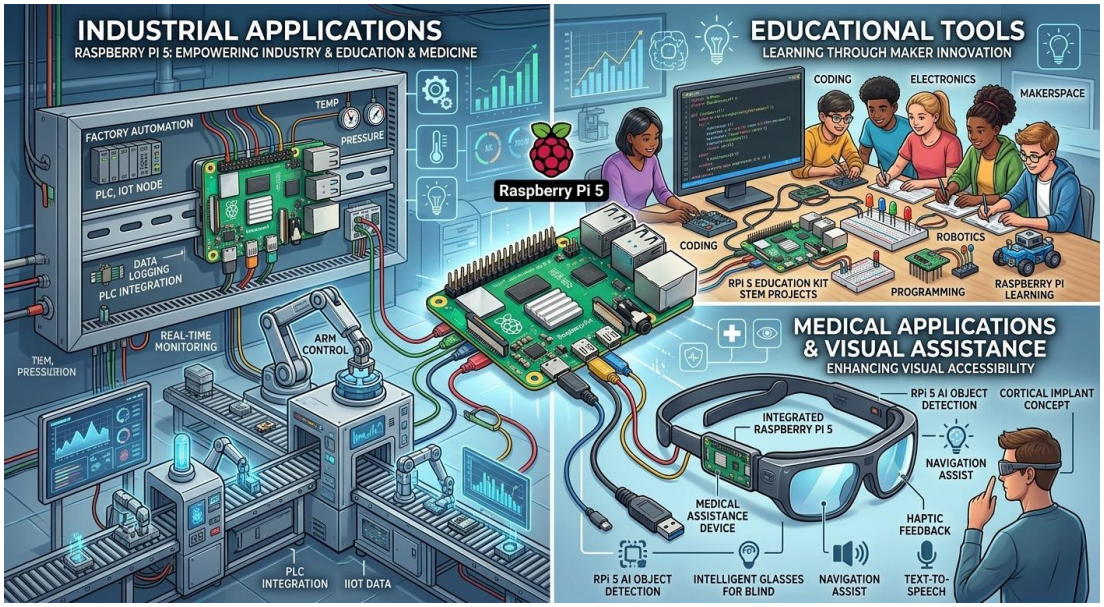
Capstone Big Idea:

Integrating AI-driven edge computing with the Internet of Things (IoT) for next-generation industrial automation, educational robotics, or assistive medical technologies is the main objective of this year's challenge. Students must design a system that analyzes the current situation—including the necessary sensing, measurements, and communication with information/data sources—to formulate AI-driven solutions and intelligent techniques aimed at expanding the industrial base, delivering medical services, or enhancing the learning environment. The proposed solution/prototype must include sensing, wireless data transfer to the cloud, advanced communication tools, and AI platforms.

Essential Question: How can integrating AI tools with information technology help develop reliable and intelligent devices that improve the industrial base, provide medical services, or enhance the learning environment?

Design Challenge:

Recently, AI has significantly improved device performance across a wide range of applications. An AI-based device can analyze and interpret variable inputs and adjust its performance accordingly. Therefore, students must design a device/system that analyzes the current situation in one of the challenges under consideration. The proposed device must perform the necessary sensing, measurements, and communication with information/data sources to support AI-driven solutions. These solutions should help expand the industrial base, deliver reliable medical services, or enhance the learning environment. The schematic diagram of the capstone idea is shown in **Scheme 1**.



Scheme 1: Schematic diagram of the capstone idea.

Pathways

The following are examples of the challenges under consideration. However, the students are not restricted to these examples.

A. Industrial Base

- Fully automated production lines.
- Product quality inspection and control.
- Any relevant application to increase the industrial base, with the approval of the school board.

B. Educational Environment

- A device that incorporates voice and text to facilitate foreign-language learning.
- A device for automatically grading homework and providing feedback to improve students' performance and support activities related to the educational process.
- A device that assists with programming in general and AI-based projects in particular.
- Any relevant device/application that facilitates the educational environment, with the approval of the school board.

C. Medical Services

- A device for people with disabilities that continuously monitors the user's condition and, in critical situations, notifies the next of kin, responsible medical personnel, and a nearby hospital. The device could also contact emergency services when immediate intervention is required.
- AI-based smart glasses that assist people who are blind with daily activities.

- An AI-based wheelchair for people with disabilities that can detect and avoid obstacles.
- Any relevant device/application to address medical problems, with the approval of the school board.

Design Requirements:

The design must meet the following requirements:

- Students must identify at least four measurable design requirements, including response time and system accuracy, for their prototype. Students must also select two additional measurable design requirements based on the application.
- The prototype must use Information and Communication Technology (ICT), IoT, and AI platforms.
- The prototype must contain hardware components.
- Students must provide datasets and a rubric for machine/deep learning algorithms during the exhibition.
- Students must provide evidence of the accuracy of the AI model, which must be at least 80%.
- Students must conduct and document sensor calibration.
- Functional prototypes must be testable, operational, reliable, and portable on exhibition day. The team should also bring a hard copy of each team member's logbook.

Constraints:

- The whole system (software and hardware) must be testable.

- Arduino boards are not allowed. Students are encouraged to use more advanced controllers.
- Students must be able to show the change in system outputs or states as a function of changing inputs.
- Laboratory equipment or materials must not be removed from the laboratories at any time. Tests that require this equipment must be conducted in the labs.
- Projects involving chronic diseases are not considered.
- For the medical pathway, no hazards are allowed.